

CHE 201 Spring 2020 HW 9

Thanh Nguyen

TOTAL POINTS

97.5 / 100

QUESTION 1

1 Question 1 12.5 / 15

- 0 pts Correct
- 2.5 pts Wrong heat transfer
- 5 pts Wrong heat transfer
- 2.5 pts Wrong entropy change
- 5 pts Wrong entropy change
- ✓ - 2.5 pts Wrong T-s diagram
- 5 pts Wrong T-s diagram

QUESTION 2

2 Question 2 15 / 15

- ✓ - 0 pts Correct

QUESTION 3

3 Question 3 15 / 15

- ✓ - 0 pts Correct

QUESTION 4

4 Question 4 15 / 15

- ✓ - 0 pts Correct
- 3 pts Wrong judgement of T2
- 3 pts Wrong judgement of S2
- 3 pts Wrong judgement of S3
- 3 pts Wrong judgement of T3
- 3 pts Wrong way of COP calculation

QUESTION 5

5 Question 5 20 / 20

- ✓ - 0 pts Correct
- 2.5 pts Wrong energy balance
- 2.5 pts Wrong electronic work
- 2.5 pts Wrong heat transfer
- 2.5 pts Wrong Tb
- 2.5 pts b) Wrong Entropy balance
- 2.5 pts b) Wrong entropy production

- 2.5 pts c) Wrong Entropy balance
- 2.5 pts c) Wrong entropy production
- 1 pts keep more significant digit

QUESTION 6

6 Question 6 20 / 20

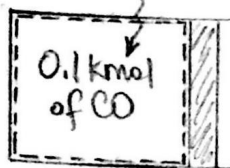
- ✓ - 0 pts Correct
- 2.5 pts Wrong final temperature
- 5 pts Wrong final temperature
- 2.5 pts Wrong final pressure
- 5 pts Wrong final pressure
- 2.5 pts Wrong entropy balance
- 5 pts Wrong entropy balance
- 2.5 pts Wrong entropy calculation
- 5 pts Wrong entropy calculation

Thanh Nguyen - CHE 201 - HW 9

1.) Sketch

$$P_1 = 150 \text{ kPa}$$

$$T_1 = 300 \text{ K}$$



$$P_2 = 500 \text{ kPa}$$

$$T_2 = 370 \text{ K}$$

Engineering Model

1. Closed system
2. Ideal gas behavior
3. $\Delta KE = \Delta PE = 0$
4. Work in the form of $p dV$

a.) For a closed system, $\Delta E = Q - W \rightarrow \Delta KE + \Delta PE + \Delta U = Q - W \rightarrow Q = \Delta U + W$
 Using table A-23, at $T_1 = 300 \text{ K} \rightarrow \bar{u}_1 = 6229 \text{ kJ/kmol}, \bar{s}_1 = 197.723 \text{ kJ/kmol}\cdot\text{K}$
 at $T_2 = 370 \text{ K} \rightarrow \bar{u}_2 = 7689 \text{ kJ/kmol}, \bar{s}_2 = 203.842 \text{ kJ/kmol}\cdot\text{K}$

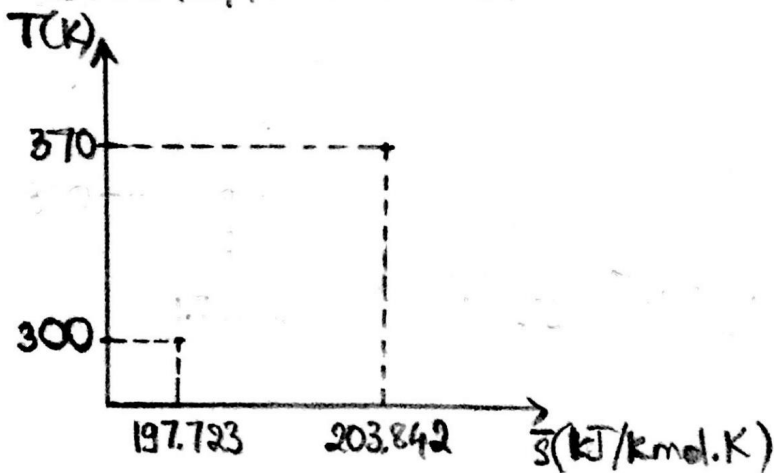
$$\rightarrow Q = \Delta U + W = n(\bar{u}_2 - \bar{u}_1) + W$$

$$= (0.1 \text{ kmol})(7689 - 6229) \text{ kJ/kmol} - 300 \text{ kJ} = -154 \text{ kJ}$$

b.) $\Delta S = n(\bar{s}_2 - \bar{s}_1 - \bar{R} \ln \frac{P_2}{P_1})$

$$= (0.1 \text{ kmol})[(203.842 - 197.723) \text{ kJ/kmol}\cdot\text{K} - 8.314 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}} \ln \frac{500}{150}]$$

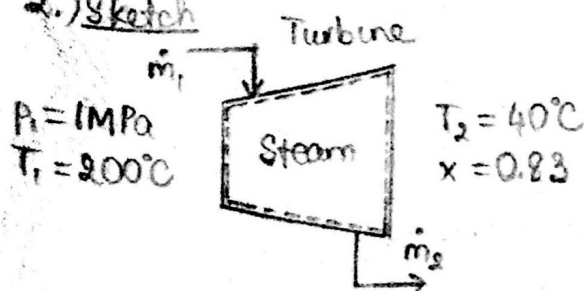
$$= -0.389 \text{ kJ/K}$$



1 Question 1 12.5 / 15

- 0 pts Correct
- 2.5 pts Wrong heat transfer
- 5 pts Wrong heat transfer
- 2.5 pts Wrong entropy change
- 5 pts Wrong entropy change
- ✓ - 2.5 pts Wrong T-s diagram
- 5 pts Wrong T-s diagram

2.) Sketch



Engineering Model

1. Control volume
2. Steady state
3. $Q = \Delta KE = \Delta PE = 0$

$$\frac{dm_{cv}}{dt} = \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\frac{dE_{cv}}{dt} = 0 \rightarrow \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + g(\cancel{z_1 - z_2}) \right] = 0$$

$$\rightarrow \dot{W}_{cv} = \dot{m}(h_1 - h_2)$$

a.) Using table A-4, at $p_1 = 1 \text{ MPa}$ and $T_1 = 200^\circ\text{C} \rightarrow h_1 = 2827.9 \text{ kJ/kg}$, $s_1 = 6.6940 \text{ kJ/kg}\cdot\text{K}$

Using table A-2, at $T_2 = 40^\circ\text{C}$

$$\rightarrow h_f = 167.57 \text{ kJ/kg}, s_f = 0.5725 \text{ kJ/kg}\cdot\text{K}$$

$$\rightarrow h_g = 2574.3 \text{ kJ/kg}, s_g = 8.2570 \text{ kJ/kg}\cdot\text{K}$$

$$\rightarrow h_2 = (1-x)h_f + xh_g = (1-0.83)(167.57 \text{ kJ/kg}) + (0.83)(2574.3 \text{ kJ/kg}) = 2165.2 \text{ kJ/kg}$$

$$\rightarrow \frac{\dot{P}}{\dot{m}} = \frac{\dot{W}_{cv}}{\dot{m}} = h_1 - h_2 = (2827.9 - 2165.2) \text{ kJ/kg} = 662.7 \text{ kJ/kg}$$

$$\text{b.) } s_2 = (1-x)s_f + xs_g = (1-0.83)(0.5725 \text{ kJ/kg}\cdot\text{K}) + (0.83)(8.2570 \text{ kJ/kg}\cdot\text{K}) = 6.9506 \text{ kJ/kg}\cdot\text{K}$$

$$\rightarrow \Delta s = s_2 - s_1 = (6.9506 - 6.6940) \text{ kJ/kg}\cdot\text{K} = 0.2566 \text{ kJ/kg}\cdot\text{K}$$

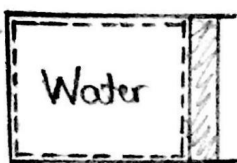
2 Question 2 15 / 15

✓ - 0 pts Correct

3.) Sketch

$$P_1 = 10 \text{ lbf/in}^2$$

$$T_1 = 500^\circ\text{F}$$



$$P_2 = 80 \text{ lbf/in}^2$$

$$T_2 = 800^\circ\text{F}$$

Engineering Model

1. Closed system
2. Internally reversible
3. Work in the form of $p dV$
4. $\Delta KE = \Delta PE = 0$

For a closed system, $\Delta E = Q - W \rightarrow \Delta KE + \Delta PE + \Delta U = Q - W \rightarrow \Delta U = Q - W$

For an internally reversible process, q is the area under the curve in $T-s$ diagram

$\rightarrow$ In this case, Q is the area of the trapezoid

$$\rightarrow q = \frac{1}{2} (T_1 + T_2) (s_2 - s_1)$$

Using table A-4

$$\rightarrow \text{at } P_1 = 10 \text{ lbf/in}^2, T_1 = 500^\circ\text{F} \rightarrow u_1 = 1182.2 \text{ Btu/lb}, s_1 = 1.9690 \text{ Btu/lb}^\circ\text{R}$$

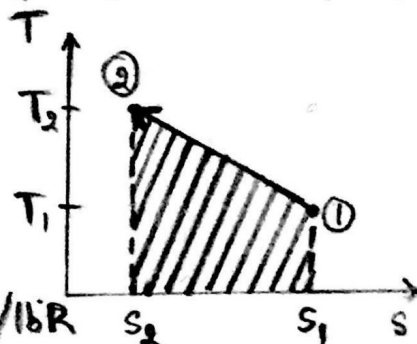
$$\rightarrow \text{at } P_2 = 80 \text{ lbf/in}^2, T_2 = 800^\circ\text{F} \rightarrow u_2 = 1292.4 \text{ Btu/lb}, s_2 = 1.8700 \text{ Btu/lb}^\circ\text{R}$$

$$\rightarrow T_1 = 500^\circ\text{F} + 460 = 960^\circ\text{R}; T_2 = 800^\circ\text{F} + 460 = 1260^\circ\text{R}$$

$$\rightarrow q = \frac{1}{2} (960 + 1260)^\circ\text{R} (1.8700 - 1.9690) \text{ Btu/lb}^\circ\text{R} = -109.89 \text{ Btu/lb}$$

$$\Delta U = Q - W \rightarrow w = q - \Delta u = -109.89 \text{ Btu/lb} - (1292.4 - 1182.2) \text{ Btu/lb}$$

$$= -220.09 \text{ Btu/lb}$$



4.)

Using table A-10, at $T_2 = 0^\circ\text{C} \rightarrow s_2 = s_g = 0.9190 \text{ kJ/kg}^\circ\text{K}$

$$\rightarrow s_3 = s_2 = 0.9190 \text{ kJ/kg}^\circ\text{K}$$

Using table A-12, at $P_3 = 16 \text{ bar}$

$$\rightarrow T_a = 60^\circ\text{C} \rightarrow s_a = 0.9069 \text{ kJ/kg}^\circ\text{K}$$

$$\rightarrow T_b = 70^\circ\text{C} \rightarrow s_b = 0.9457 \text{ kJ/kg}^\circ\text{K}$$

$$\rightarrow \text{Using interpolation: } \frac{T_3 - T_a}{T_b - T_a} = \frac{s_3 - s_a}{s_b - s_a} \rightarrow \frac{T_3 - 60^\circ\text{C}}{70^\circ\text{C} - 60^\circ\text{C}} = \frac{(0.9190 - 0.9069) \text{ kJ/kg}^\circ\text{K}}{(0.9457 - 0.9069) \text{ kJ/kg}^\circ\text{K}}$$

$$\rightarrow T_3 = 63.12^\circ\text{C}$$

$$T_c = T_2 = 0^\circ\text{C} + 273 = 273 \text{ K}$$

$$T_4 = T_3 = 63.12^\circ\text{C} + 273 = 336.12 \text{ K}$$

$$\text{Since this is a Carnot cycle} \rightarrow \beta = \frac{T_c}{T_4 - T_c} = \frac{273 \text{ K}}{(336.12 - 273) \text{ K}} = 4.33$$

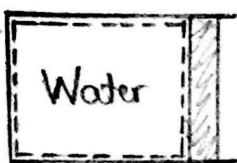
3 Question 3 15 / 15

✓ - 0 pts Correct

3.) Sketch

$$P_1 = 10 \text{ lbf/in}^2$$

$$T_1 = 500^\circ\text{F}$$



$$P_2 = 80 \text{ lbf/in}^2$$

$$T_2 = 800^\circ\text{F}$$

Engineering Model

1. Closed system
2. Internally reversible
3. Work in the form of $p dv$
4. $\Delta KE = \Delta PE = 0$

For a closed system, $\Delta E = Q - W \rightarrow \Delta KE + \Delta PE + \Delta U = Q - W \rightarrow \Delta U = Q - W$

For an internally reversible process, q is the area under the curve in $T-s$ diagram

$\rightarrow$ In this case, Q is the area of the trapezoid

$$\rightarrow q = \frac{1}{2} (T_1 + T_2) (s_2 - s_1)$$

Using table A-4

$$\rightarrow \text{at } P_1 = 10 \text{ lbf/in}^2, T_1 = 500^\circ\text{F} \rightarrow u_1 = 1182.2 \text{ Btu/lb}, s_1 = 1.9690 \text{ Btu/lb}^\circ\text{R}$$

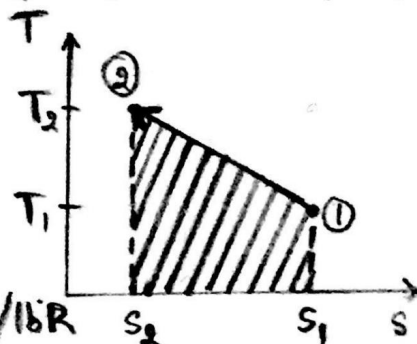
$$\rightarrow \text{at } P_2 = 80 \text{ lbf/in}^2, T_2 = 800^\circ\text{F} \rightarrow u_2 = 1292.4 \text{ Btu/lb}, s_2 = 1.8700 \text{ Btu/lb}^\circ\text{R}$$

$$\rightarrow T_1 = 500^\circ\text{F} + 460 = 960^\circ\text{R}; T_2 = 800^\circ\text{F} + 460 = 1260^\circ\text{R}$$

$$\rightarrow q = \frac{1}{2} (960 + 1260)^\circ\text{R} (1.8700 - 1.9690) \text{ Btu/lb}^\circ\text{R} = -109.89 \text{ Btu/lb}$$

$$\Delta U = Q - W \rightarrow w = q - \Delta u = -109.89 \text{ Btu/lb} - (1292.4 - 1182.2) \text{ Btu/lb}$$

$$= -220.09 \text{ Btu/lb}$$



4.)

Using table A-10, at $T_2 = 0^\circ\text{C} \rightarrow s_2 = s_g = 0.9190 \text{ kJ/kg}^\circ\text{K}$

$$\rightarrow s_3 = s_2 = 0.9190 \text{ kJ/kg}^\circ\text{K}$$

Using table A-12, at $P_3 = 16 \text{ bar}$

$$\rightarrow T_a = 60^\circ\text{C} \rightarrow s_a = 0.9069 \text{ kJ/kg}^\circ\text{K}$$

$$\rightarrow T_b = 70^\circ\text{C} \rightarrow s_b = 0.9457 \text{ kJ/kg}^\circ\text{K}$$

$$\rightarrow \text{Using interpolation: } \frac{T_3 - T_a}{T_b - T_a} = \frac{s_3 - s_a}{s_b - s_a} \rightarrow \frac{T_3 - 60^\circ\text{C}}{70^\circ\text{C} - 60^\circ\text{C}} = \frac{(0.9190 - 0.9069) \text{ kJ/kg}^\circ\text{K}}{(0.9457 - 0.9069) \text{ kJ/kg}^\circ\text{K}}$$

$$\rightarrow T_3 = 63.12^\circ\text{C}$$

$$T_c = T_2 = 0^\circ\text{C} + 273 = 273 \text{ K}$$

$$T_4 = T_3 = 63.12^\circ\text{C} + 273 = 336.12 \text{ K}$$

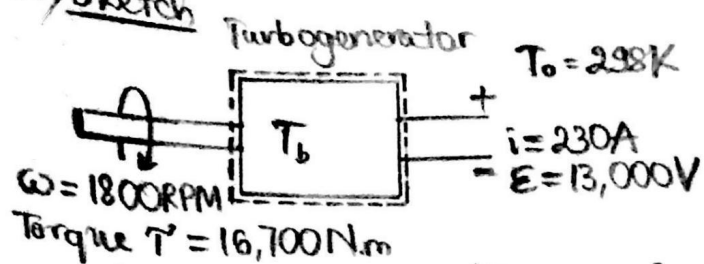
$$\text{Since this is a Carnot cycle} \rightarrow \beta = \frac{T_c}{T_4 - T_c} = \frac{273 \text{ K}}{(336.12 - 273) \text{ K}} = 4.33$$

4 Question 4 15 / 15

✓ - 0 pts Correct

- 3 pts Wrong judgement of T2
- 3 pts Wrong judgement of S2
- 3 pts Wrong judgement of S3
- 3 pts Wrong judgement of T3
- 3 pts Wrong way of COP calculation

5.) Sketch



Engineering Model

1. Closed system
2. $\Delta KE = \Delta PE = 0$
3. Steady state

a.) For a closed system, $\frac{dE}{dt} = 0 \rightarrow \dot{Q} - \dot{W} = 0 \rightarrow \dot{Q} = \dot{W}_{\text{shaft}} + \dot{W}_{\text{elec}}$

$$\dot{W}_{\text{shaft}} = -T\omega = -(16,700\text{Nm})(1800\text{RPM})\left(\frac{2\pi\text{rad}}{1\text{rev}}\right)\left(\frac{1\text{min}}{60\text{s}}\right)\left(\frac{1\text{kJ}}{1000\text{Nm}}\right)\left(\frac{1\text{KW}}{1\text{kJ/s}}\right) = -3148\text{KW}$$

$$\dot{W}_{\text{elec}} = EI = (13,000\text{V})(230\text{A}) = 2.99 \times 10^6\text{W} = 2990\text{KW}$$

$$\rightarrow \dot{Q} = \dot{W}_{\text{shaft}} + \dot{W}_{\text{elec}} = -3148\text{KW} + 2990\text{KW} = -158\text{KW}$$

$$\dot{Q} = -hA(T_b - T_0) \rightarrow T_b = T_0 - \frac{\dot{Q}}{hA} = 298\text{K} + \frac{158\text{KW}}{(110\text{W/m}^2\text{K})(32\text{m}^2)} \times \frac{1000\text{W}}{1\text{KW}} = 342.9\text{K}$$

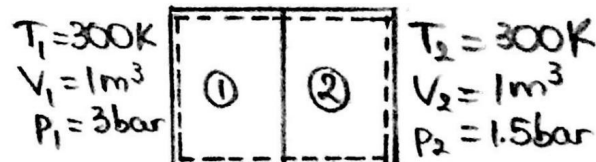
b.) At steady state, $\Delta S = 0$

$$\Delta S = \int \frac{\delta Q}{T} + \sigma \rightarrow \sigma = -\int \frac{\delta Q}{T} = -\frac{Q}{T_b} = -\frac{-158\text{KW}}{342.9\text{K}} = 0.4608\text{KW/K}$$

c.) Using the above equation $\sigma = -\frac{Q}{T_0} = -\frac{-158\text{KW}}{298\text{K}} = 0.5302\text{KW/K}$

6.) Engineering Model

1. Closed system
2. Ideal gas behavior
3. No work is done $W = 0$
4. $Q = \Delta KE = \Delta PE = 0$



a.) For a closed system $\Delta U = Q - W \rightarrow \Delta U = 0 \rightarrow \Delta U_1 + \Delta U_2 = 0$

$$\rightarrow m_1(u_f - u_1) + m_2(u_f - u_2) = 0 \rightarrow (m_1 + m_2)(u_f - u_1) = 0 \quad (\text{since } u_1 = u_2 \text{ at } T = 300\text{K})$$

$$\rightarrow u_f = u_1 \rightarrow T_f = T_1 = T_2 = 300\text{K} \quad ; \quad V_1 = V_2 = V \text{ and } V_f = 2V$$

$$\text{b.) } m_1 = \frac{P_1 V_1}{RT_1} = \frac{P_1 V}{RT} \quad ; \quad m_2 = \frac{P_2 V_2}{RT_2} = \frac{P_2 V}{RT} \quad ; \quad m_1 + m_2 = m_f = \frac{P_f V_f}{RT_f} = \frac{2P_f V}{RT}$$

$$\rightarrow \frac{P_1 V}{RT} + \frac{P_2 V}{RT} = \frac{2P_f V}{RT} \rightarrow P_f = \frac{P_1 + P_2}{2} = \frac{3\text{bar} + 1.5\text{bar}}{2} = 2.25\text{bar}$$

c.) Since $Q = 0 \rightarrow \int \frac{\delta Q}{T} = 0 \rightarrow \Delta S = \sigma \rightarrow \sigma = \Delta S_1 + \Delta S_2$

$$\text{since } T_1 = T_2 = T_f \rightarrow s_1^0 = s_2^0 = s_f^0$$

$$\rightarrow \sigma = m_1(s_f^0 - s_1^0 - R \ln \frac{P_f}{P_1}) + m_2(s_f^0 - s_2^0 - R \ln \frac{P_f}{P_2}) = -R(m_1 \ln \frac{P_f}{P_1} + m_2 \ln \frac{P_f}{P_2})$$

$$= -R\left(\frac{P_1 V}{RT} \ln \frac{P_f}{P_1} + \frac{P_2 V}{RT} \ln \frac{P_f}{P_2}\right) = -\frac{V}{T}\left(P_1 \ln \frac{P_f}{P_1} + P_2 \ln \frac{P_f}{P_2}\right)$$

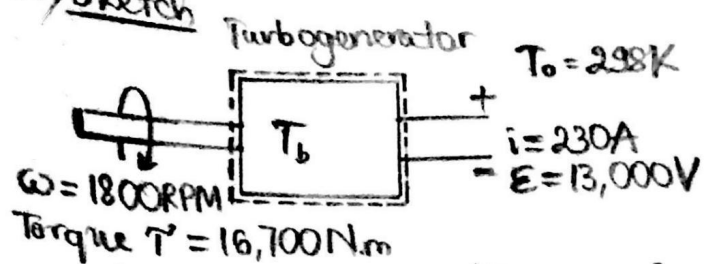
$$= -\frac{1\text{m}^3}{300\text{K}} \left[(3\text{bar}) \ln \frac{2.25}{3} + (1.5\text{bar}) \ln \frac{2.25}{1.5}\right] \times \frac{10^5\text{N/m}^2}{1\text{bar}} \times \frac{1\text{kJ}}{10^3\text{Nm}} = 0.0849\text{KJ/K}$$

5 Question 5 20 / 20

✓ - 0 pts Correct

- 2.5 pts Wrong energy balance
- 2.5 pts Wrong electronic work
- 2.5 pts Wrong heat transfer
- 2.5 pts Wrong T_b
- 2.5 pts b) Wrong Entropy balance
- 2.5 pts b) Wrong entropy production
- 2.5 pts c) Wrong Entropy balance
- 2.5 pts c) Wrong entropy production
- 1 pts keep more significant digit

5.) Sketch



Engineering Model

1. Closed system
2. $\Delta KE = \Delta PE = 0$
3. Steady state

a.) For a closed system, $\frac{dE}{dt} = 0 \rightarrow \dot{Q} - \dot{W} = 0 \rightarrow \dot{Q} = \dot{W}_{shaft} + \dot{W}_{elec}$

$$\dot{W}_{shaft} = -T\omega = -(16,700Nm)(1800RPM)\left(\frac{2\pi rad}{1 rev}\right)\left(\frac{1 min}{60s}\right)\left(\frac{1 kJ}{1000 Nm}\right)\left(\frac{1 kW}{1 kJ/s}\right) = -3148 kW$$

$$\dot{W}_{elec} = EI = (13,000V)(230A) = 2.99 \times 10^6 W = 2990 kW$$

$$\rightarrow \dot{Q} = \dot{W}_{shaft} + \dot{W}_{elec} = -3148 kW + 2990 kW = -158 kW$$

$$\dot{Q} = -hA(T_b - T_0) \rightarrow T_b = T_0 - \frac{\dot{Q}}{hA} = 298K + \frac{158 kW}{(110 W/m^2K)(32 m^2)} \times \frac{1000 W}{1 kW} = 342.9 K$$

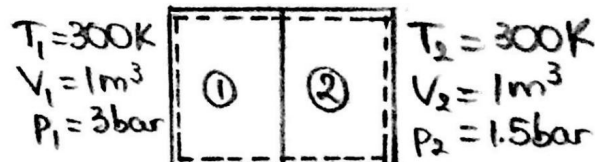
b.) At steady state, $\Delta S = 0$

$$\Delta S = \int \frac{\delta Q}{T} + \sigma \rightarrow \sigma = - \int \frac{\delta Q}{T} = - \frac{Q}{T_b} = - \frac{-158 kW}{342.9 K} = 0.4608 kW/K$$

c.) Using the above equation $\sigma = - \frac{Q}{T_0} = - \frac{-158 kW}{298 K} = 0.5302 kW/K$

6.) Engineering Model

1. Closed system
2. Ideal gas behavior
3. No work is done $W = 0$
4. $Q = \Delta KE = \Delta PE = 0$



a.) For a closed system $\Delta U = Q - W \rightarrow \Delta U = 0 \rightarrow \Delta U_1 + \Delta U_2 = 0$

$$\rightarrow m_1(u_f - u_1) + m_2(u_f - u_2) = 0 \rightarrow (m_1 + m_2)(u_f - u_1) = 0 \text{ (since } u_1 = u_2 \text{ at } T = 300K)$$

$$\rightarrow u_f = u_1 \rightarrow T_f = T_1 = T_2 = 300K \quad ; \quad V_1 = V_2 = V \text{ and } V_f = 2V$$

$$b.) m_1 = \frac{P_1 V_1}{RT_1} = \frac{P_1 V}{RT} ; m_2 = \frac{P_2 V_2}{RT_2} = \frac{P_2 V}{RT} ; m_1 + m_2 = m_f = \frac{P_f V_f}{RT_f} = \frac{2P_f V}{RT}$$

$$\rightarrow \frac{P_1 V}{RT} + \frac{P_2 V}{RT} = \frac{2P_f V}{RT} \rightarrow P_f = \frac{P_1 + P_2}{2} = \frac{3bar + 1.5bar}{2} = 2.25 bar$$

c.) Since $Q = 0 \rightarrow \int \frac{\delta Q}{T} = 0 \rightarrow \Delta S = \sigma \rightarrow \sigma = \Delta S_1 + \Delta S_2$

$$\text{since } T_1 = T_2 = T_f \rightarrow s_1^o = s_2^o = s_f^o$$

$$\rightarrow \sigma = m_1(s_f^o - s_1^o - R \ln \frac{P_f}{P_1}) + m_2(s_f^o - s_2^o - R \ln \frac{P_f}{P_2}) = -R(m_1 \ln \frac{P_f}{P_1} + m_2 \ln \frac{P_f}{P_2})$$

$$= -R \left(\frac{P_1 V}{RT} \ln \frac{P_f}{P_1} + \frac{P_2 V}{RT} \ln \frac{P_f}{P_2} \right) = - \frac{V}{T} \left(P_1 \ln \frac{P_f}{P_1} + P_2 \ln \frac{P_f}{P_2} \right)$$

$$= - \frac{1m^3}{300K} \left[(3bar) \ln \frac{2.25}{3} + (1.5bar) \ln \frac{2.25}{1.5} \right] \times \frac{10^5 N/m^2}{1bar} \times \frac{1 kJ}{10^3 Nm} = 0.0849 kJ/K$$

6 Question 6 20 / 20

✓ - 0 pts Correct

- 2.5 pts Wrong final temperature
- 5 pts Wrong final temperature
- 2.5 pts Wrong final pressure
- 5 pts Wrong final pressure
- 2.5 pts Wrong entropy balance
- 5 pts Wrong entropy balance
- 2.5 pts Wrong entropy calculation
- 5 pts Wrong entropy calculation